

Convergent Privacy Framework for Multi-layer GNNs through Contractive Message Passing



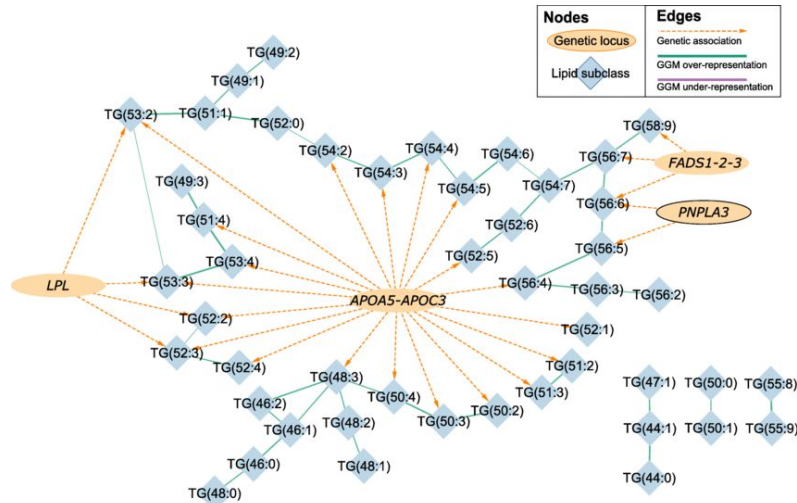
Yu Zheng*, Chenang Li*, Zhou Li*,
Qingsong Wang[#]



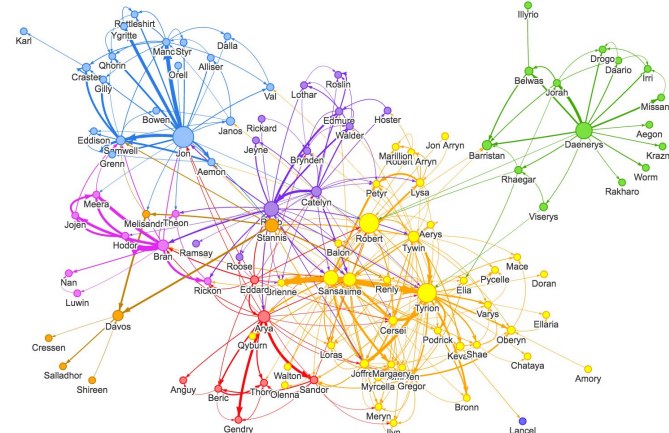
*** University of California, Irvine**
University of California, San Diego



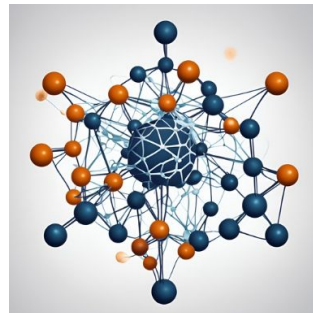
Graphs are Everywhere



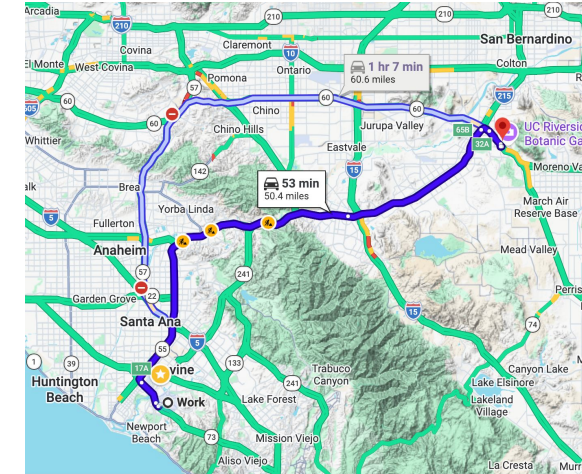
Genetic association



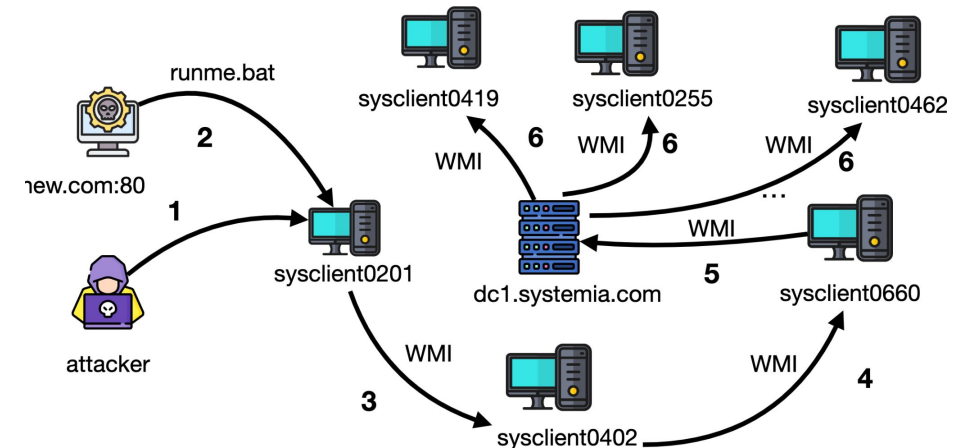
Social Networks



Chemical/Drug Molecules



Maps

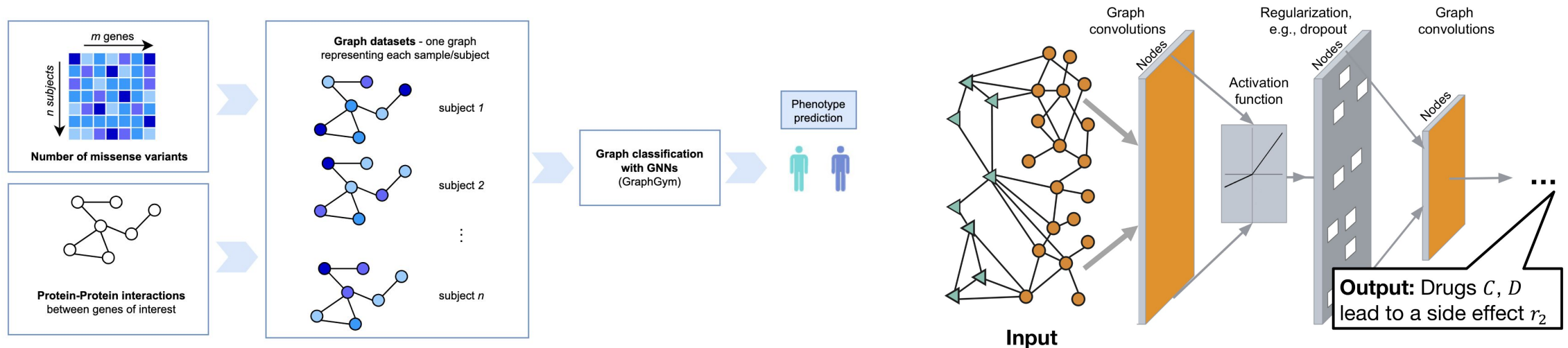


Network Logs

Graph
 $G = (V, E)$

Graph Neural Networks

Graph Neural Networks (GNNs) are designed for structural data.



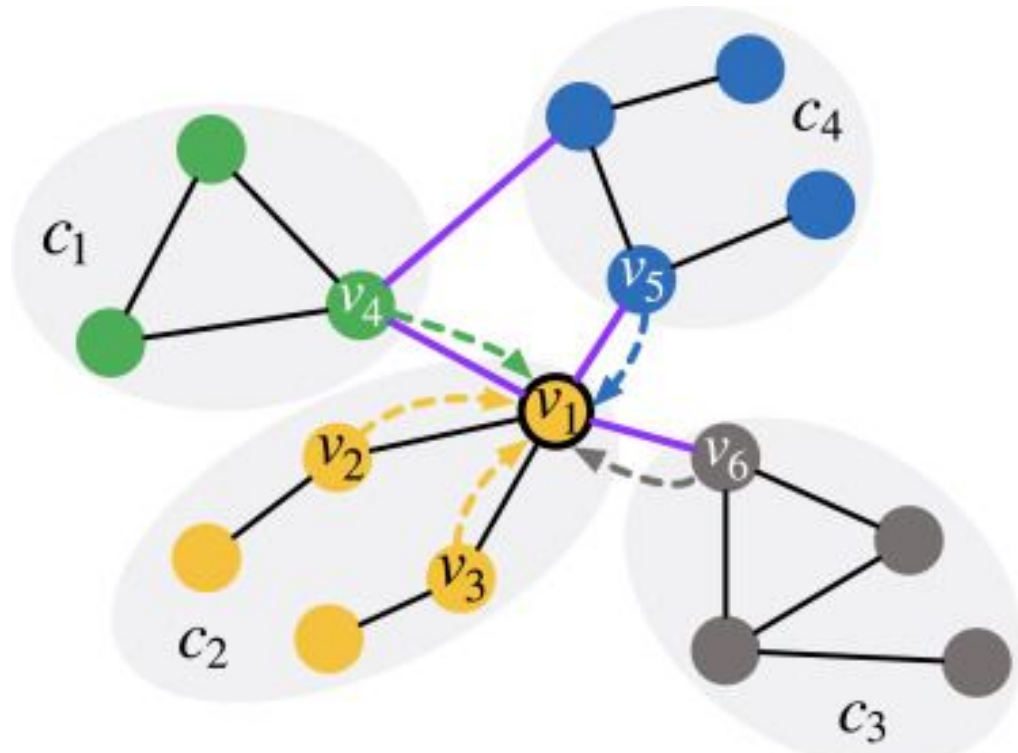
Early diagnosis of Alzheimer's disease

Drug analysis of side effects

[1] L. Hernández-Lorenzo. On the limits of graph neural networks for the early diagnosis of Alzheimer's disease. In Scientific Reports, 12(1), 17632.

[2] J. Leskovec. Graph Neural Networks for Multirelational Link Prediction. In CS224W: Machine Learning with Graphs.

Message-Passing GNNs

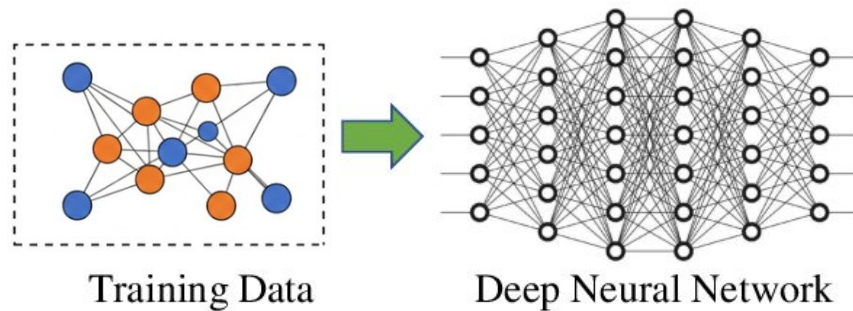


- Message-passing paradigm: aggregating information from their neighbors.
 - v_1 : aggregated from neighbors.
- Aggregation layer by layer.

[2] Enhancing Graph Neural Networks by a High-quality Aggregation of Beneficial Information. Neural Networks, 2021.

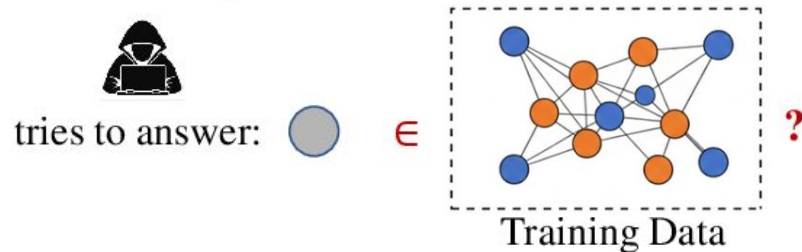
Privacy Issue in GNNs

Training of Target Model



- Membership inference attacks (MIAs)
 - Node: is **this user** in the training social network?
 - Edge: is there **a link** between user A and B in training?
 - Subgraph: was this **community structure** used?

Membership Inference Attack on Target Model

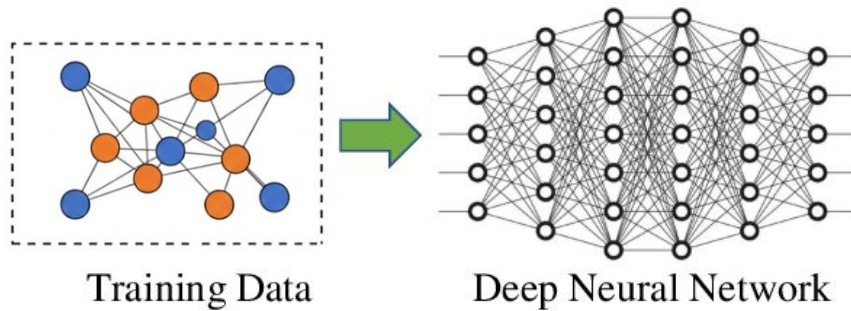


[1] F. Wu, Y. Long, C. Zhang, and B. Li, "LINKTELLER: recovering private edges from graph neural networks via influence analysis. IEEE S&P, 2022.

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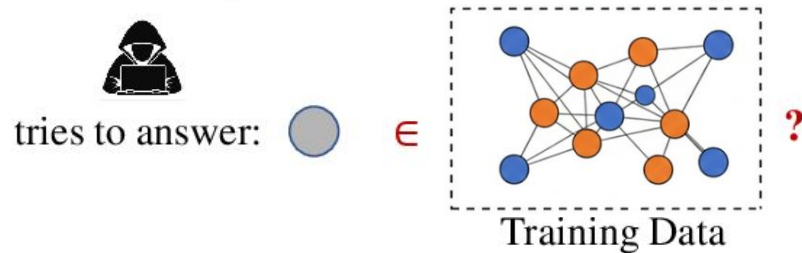
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Membership Inference Attack on Target Model



Privacy attacks, e.g., MIAs, can recover sensitive information encoded within the graph via *black-box* model access.

[1] F. Wu, Y. Long, C. Zhang, and B. Li, "LINKTELLER: recovering private edges from graph neural networks via influence analysis. IEEE S&P, 2022.

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Differential Privacy (DP) for Graphs

- Definition 1. Two datasets D , D' are adjacent if they differ by only one data instance. A random mechanism M is (ϵ, δ) -differentially private if for all adjacent datasets D , D' and for all events S in the output space of M , we have

$$\Pr(M(D) \in S) \leq e^\epsilon \Pr(M(D') \in S) + \delta.$$

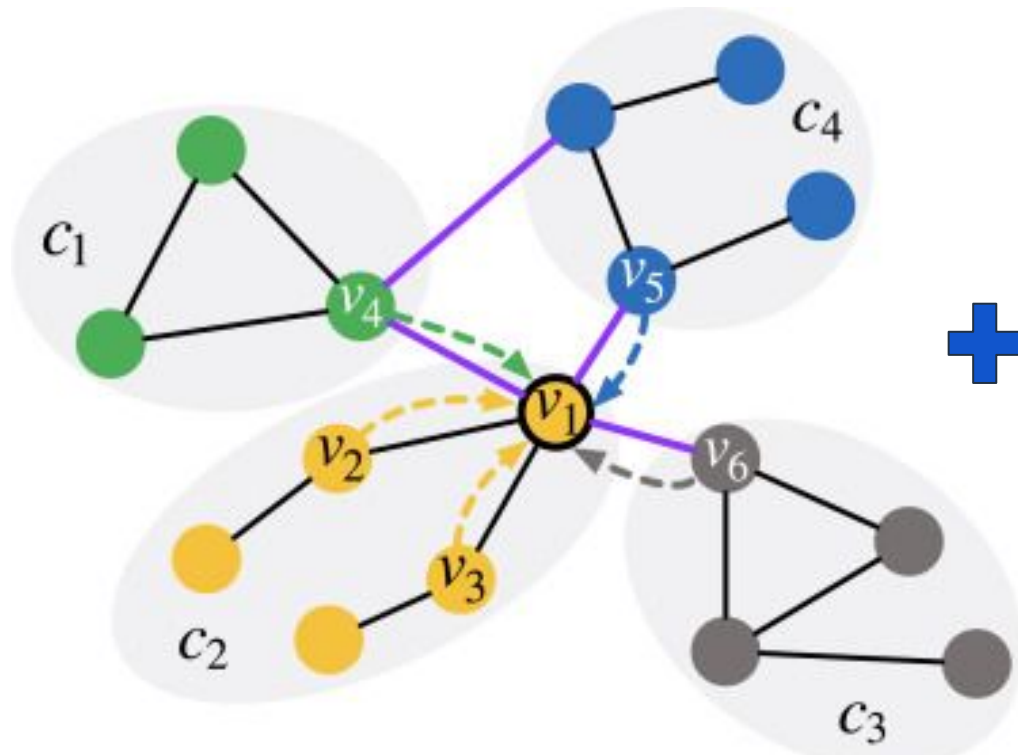
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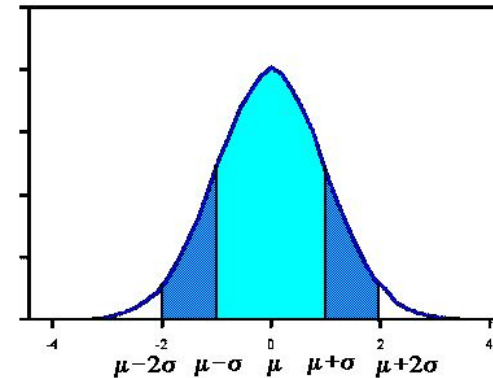
$$\Pr(M(D) \in S) \leq e^\epsilon \Pr(M(D') \in S) + \delta.$$

- Edge-level neighboring & node-level neighboring.

Perturbed Message-Passing GNNs



+



Add Gaussian noise.

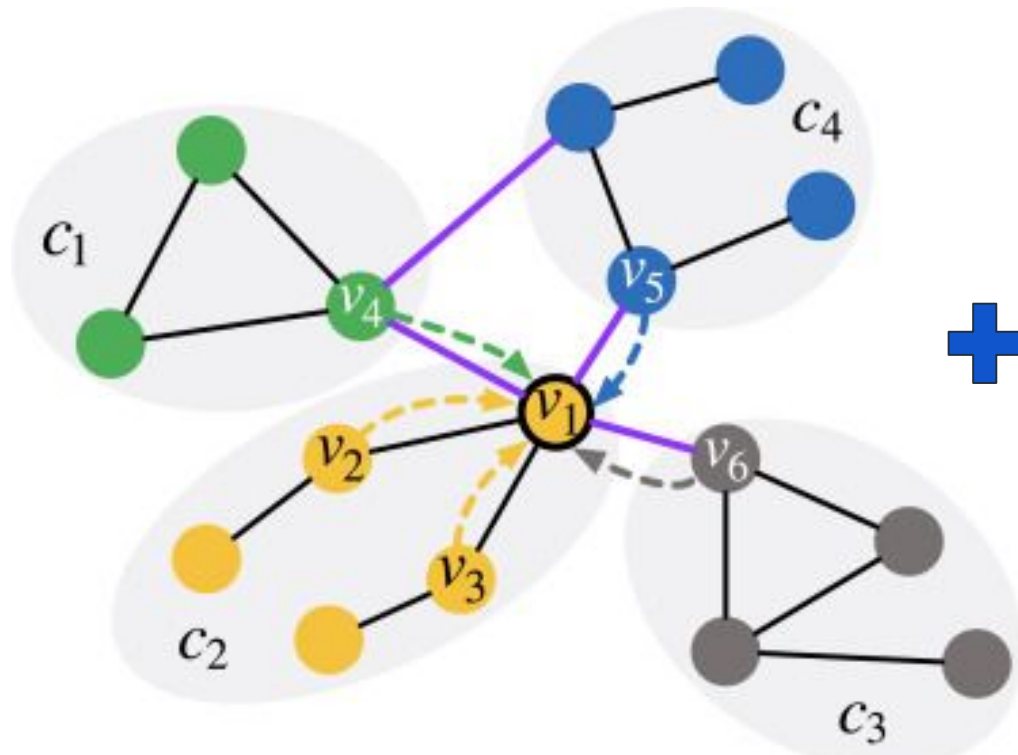


V_1

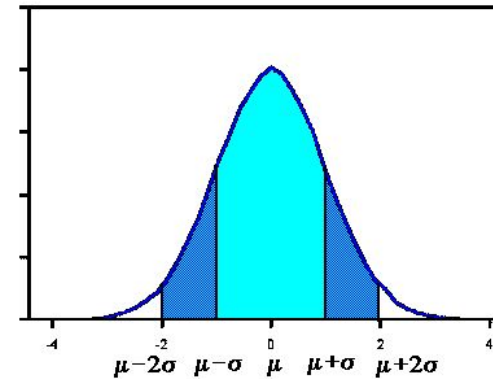
V_1'

Output node representations are similar.

Perturbed Message-Passing GNNs



+



Add Gaussian
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V_1

V_1'

Output node
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Perturbed Message-Passing GNNs

- **Formulation on perturbed message passing:**

$$\mathbf{X}^{(k+1)} = \Pi(\text{MP}_G(\mathbf{X}^{(k)}) + \mathbf{Z}^{(k)})$$



Node representation/embeddings;
 $\mathbf{X}^{(0)}$: input feature matrix.



Gaussian noise $\sim \mathcal{N}(0, \sigma^2)$.

State-of-the-Arts

TABLE I: Comparison between Private GNNs. EDP and NDP summarizes the results of private GNNs in Table III.

Framework	Mechanism	Complexity per Layer	Calibrated Noise (σ)	EDP Utility	NDP Utility
PertGraph [45, 26]	Graph perturbation	$O(V ^2)$	$\propto 1$	★ ★ ☆ ☆ ☆	★ ☆ ☆ ☆ ☆
DPDGC [39]	Decoupled graph with perturbation	$O(E)$	$\propto \sqrt{K}$	★ ★ ★ ☆ ☆	★ ★ ★ ★ ☆
GAP [46]	Perturbed message passing	$O(E)$	$\propto \sqrt{K}$	★ ★ ★ ★ ☆	★ ★ ★ ★ ☆
CARIBOU	Perturbed message passing	$O(E)$	$\propto \sqrt{\min(K, \frac{1-C_L^K}{1+C_L^K} \frac{1+C_L}{1-C_L})}$	★ ★ ★ ★ ★	★ ★ ★ ★ ★



- Share a critical limitation: the **privacy loss grows linearly with the number of layers K** or graph hops.
 - Require large amounts of noise to maintain a reasonable level of privacy guarantee; This, in turn, degrading model utility severely.

Motivating Scenarios & Challenges

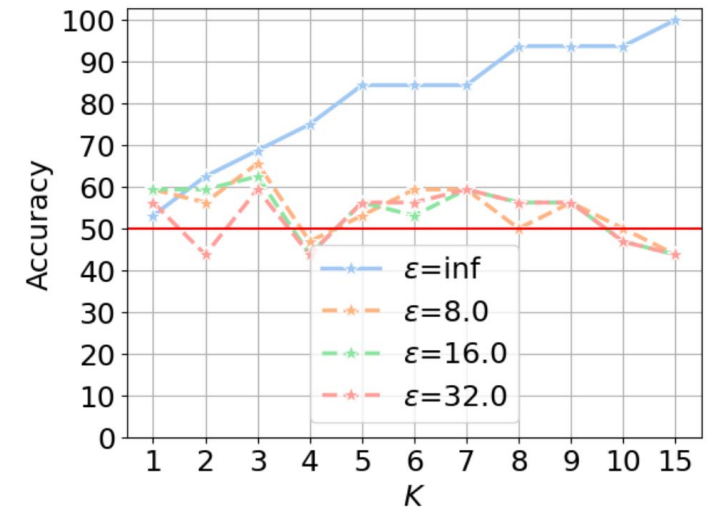
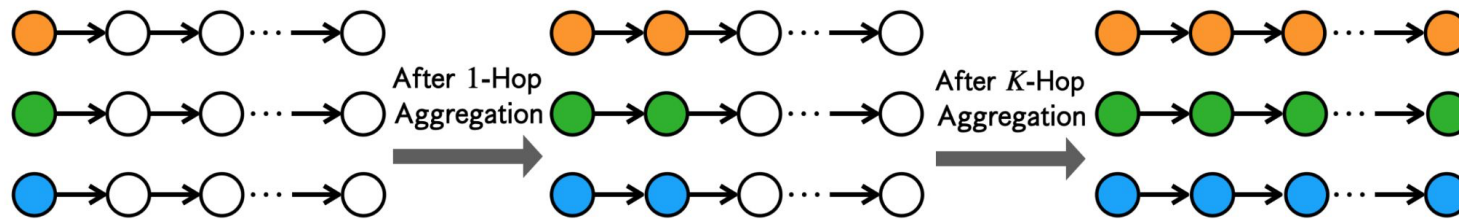
- **Multi-layer GNNs**: capture complex relations and analyze graphs with long-range interactions [3,4].
 - Example: ↑ accuracy from **72.5%** to **88.2%** [4].

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[4] Training graph neural networks with 1000 layers. ICML, 2021

Motivating Scenarios & Challenges

- **Multi-layer GNNs**: capture complex relations and analyze graphs with long-range interactions [3,4].
 - Example: ↑ accuracy from **72.5%** to **88.2%** [4].
- Challenges: larger K leads to larger privacy parameter ϵ , a.k.a weak privacy guarantee.



[3] How powerful are k-hop message passing graph neural networks. NeurIPS, 2022.

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Research Question

- “**Over-smoothing**” phenomenon [5]: node representations become increasingly homogeneous as network depth increases and consequently making membership inference more challenging.

[5] Measuring and relieving the over-smoothing problem for graph neural networks from the topological view. AAAI, 2020.

Research Question

- “**Over-smoothing**” phenomenon [5]: node representations become increasingly homogeneous as network depth increases and consequently making membership inference more challenging.



Motivate

- Can we achieve differentially private graph learning with a **convergent (bounded)** privacy budget, thereby improving the privacy-utility trade-off for deeper GNNs?
 - **Not linearly increase with K !**

[5] Measuring and relieving the over-smoothing problem for graph neural networks from the topological view. AAAI, 2020.

Core Idea for Convergent Privacy

- Insight: leverage the inherent **privacy amplification** that occurs in multi-layer GNNs through **contractiveness**.
 - Motivated by DP-GD: **converge** to a finite value with **arbitrarily** many iterations.

Core Idea for Convergent Privacy

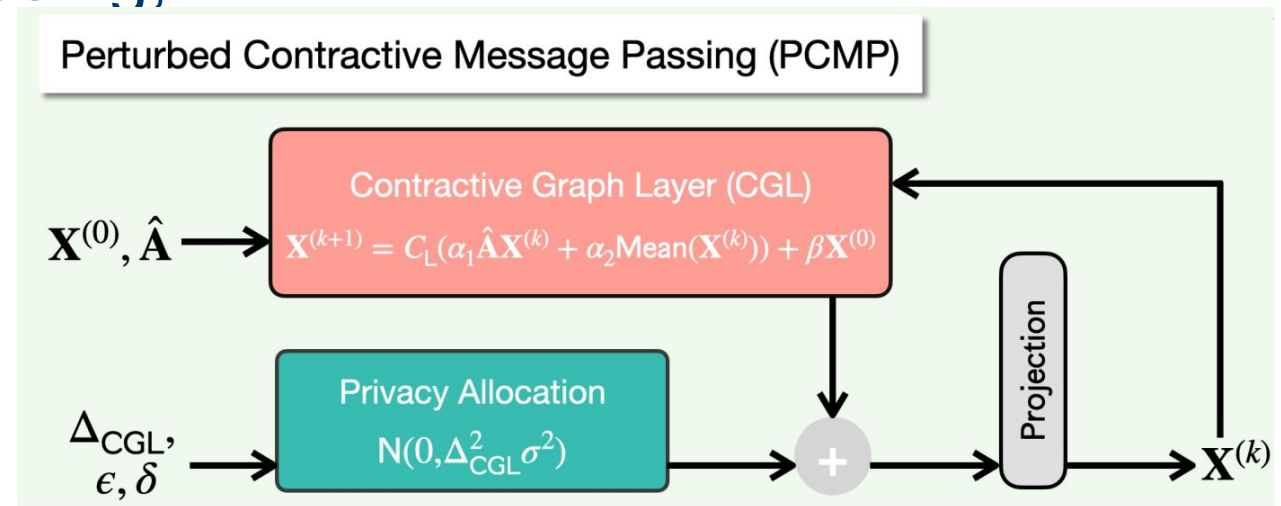
- Insight: leverage the inherent **privacy amplification** that occurs in multi-layer GNNs through **contractiveness**.
 - Motivated by DP-GD: **converge** to a finite value with **arbitrarily** many iterations.
- When perturbed message passing is contractive, the **distance** between GNNs trained on neighboring datasets **shrinks** at each step.
 - *Translate* the advanced privacy analysis techniques from DP-GD to GNNs.

Core Idea for Convergent Privacy

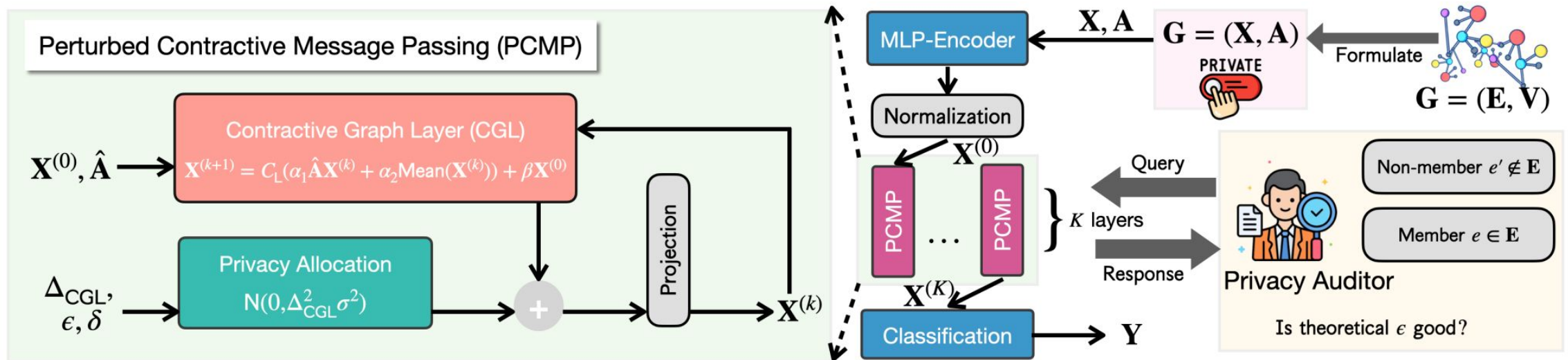
- Consequently, the influence of individual data points diminishes, leading to the amplified privacy rooted from “over-smoothing”.
 - Accordingly, remove the over-estimated privacy loss;
 - Derive a much tighter bound for finally released GNN model.

Core Idea for Convergent Privacy

- Consequently, the influence of individual data points diminishes, leading to the **amplified privacy** rooted from “**over-smoothing**”.
 - Accordingly, remove the over-estimated privacy loss;
 - Derive a much tighter bound for finally released GNN model.
- Two critical conditions:
 - Contractive message passing;
 - Release only $\mathbf{X}^{(k)}$.

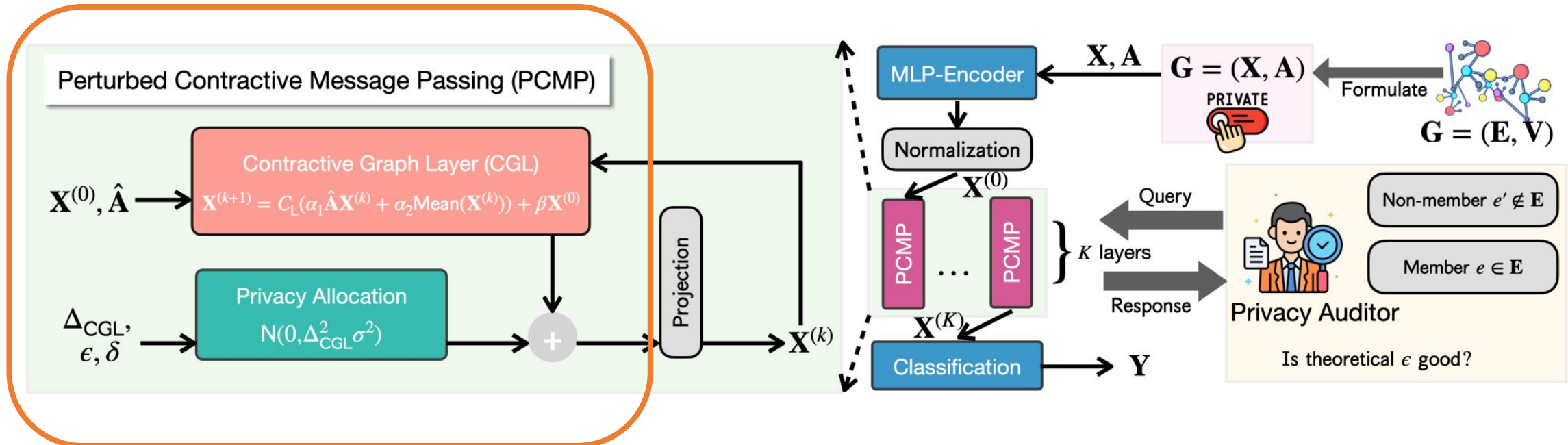


Overview of Caribou



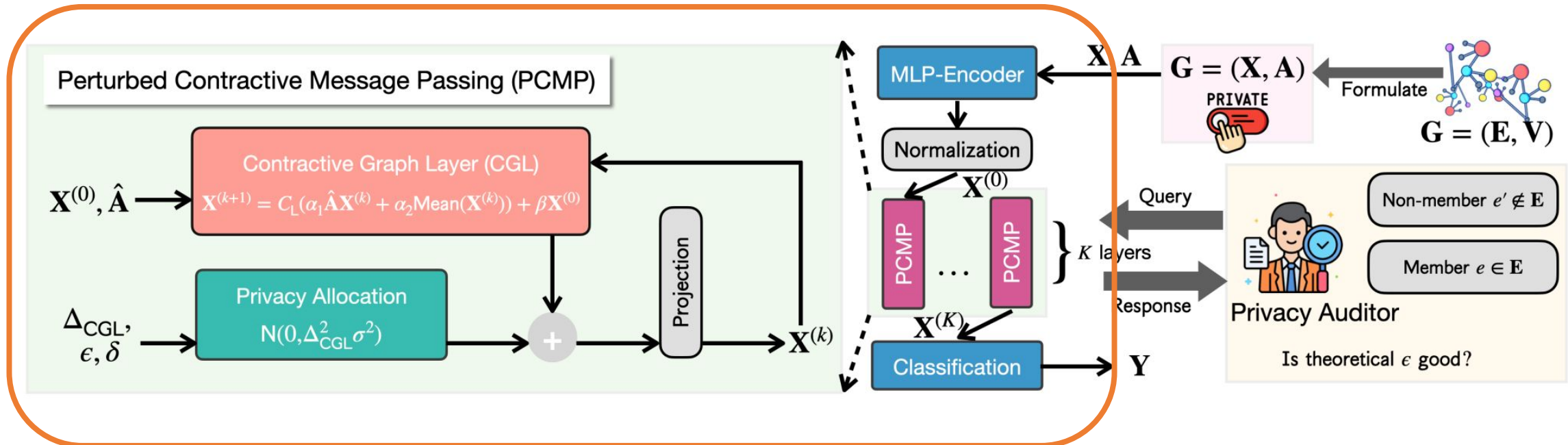
Overview of Caribou

- **C**ontractive **A**ggregation **M**odule (CAM).
- **P**rivacy **A**llocation **M**odule (PAM).



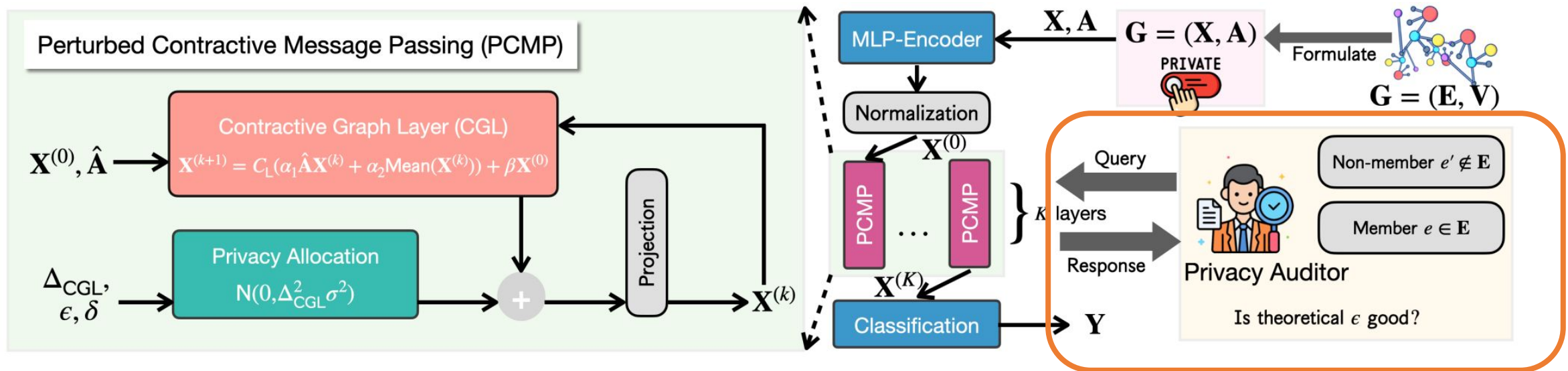
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- **P**rivacy **A**uditing **M**odule (PDM).



Core Algorithm

- Step 1. Calculate Sensitivity;

```

1:
2:  ▷ Calculate the Required Noise Calibration (from PAM).
3: if Edge-level privacy then
4:     Calculate  $\Delta(\text{CGL})$  through Equation 6
5: else if Node-level privacy then
6:     Calculate  $\Delta(\text{CGL})$  through Equation 8
7: end if
8: Calculate  $\sigma^2$  through Theorem 6
9:
10:  ▷ Perturbed Contractive Message Passing (from CAM).
11: for  $k = 0, \dots, K - 1$  do
12:      $\mathbf{X}^{(k+1)} \leftarrow C_L(\alpha_1 \hat{A} \mathbf{X}^{(k)} + \alpha_2 \text{Mean}(\mathbf{X}^{(k)})) + \beta \mathbf{X}^{(0)}$ 
13:     ▷ Contractive graph layer: compute node embeddings.
14:      $\mathbf{X}^{(k+1)} \leftarrow \mathbf{X}^{(k+1)} + \mathcal{N}(\mu, (\Delta(\text{CGL}))^2 \sigma^2)$ 
15:                                     ▷ DP Perturbation.
16:      $\mathbf{X}^{(k+1)} \leftarrow \Pi_{\mathcal{K}}(\mathbf{X}^{(k)})$       ▷ Projection with norm 1.
17: end for
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19: Return:  $\mathbf{X}^{(K)}$ 

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- Step 1. Calculate sensitivity;
- Step 2. Contractive message passing;
- Step 3. Add appropriate DP noise;

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-
- Repeat Steps 1-4 for all layers.
 - Return the final node representation.

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- Step 3. Add appropriate DP noise;
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Algorithm 1: Private Multi-hop Aggregation

Input : Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with adjacency matrix $\mathbf{A}$; initial normalized features $\tilde{\mathbf{X}}^{(0)}$; max hop K ; noise variance σ^2 ;
Output : Private aggregated node feature matrices $\tilde{\mathbf{X}}^{(1)}, \dots, \tilde{\mathbf{X}}^{(K)}$

```

1 for  $k \in \{1, \dots, K\}$  do
2    $\mathbf{X}^{(k)} \leftarrow \mathbf{A}^T \cdot \tilde{\mathbf{X}}^{(k-1)}$            // aggregate
3    $\tilde{\mathbf{X}}^{(k)} \leftarrow \mathbf{X}^{(k)} + \mathcal{N}(\sigma^2 \mathbb{I})$        // perturb
4   for  $v \in \mathcal{V}$  do
5      $\tilde{\mathbf{X}}_v^{(k)} \leftarrow \tilde{\mathbf{X}}_v^{(k)} / \|\tilde{\mathbf{X}}_v^{(k)}\|_2$    // normalize
6   end
7 end
8 return  $\tilde{\mathbf{X}}^{(1)}, \dots, \tilde{\mathbf{X}}^{(K)}$ 

```

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Core Theory

Theorem 1 [DP guarantee for CGL layers]. Let $\mathbf{G}$ be a graph and K be the number of contractive graph layers in CARIBOU. Let $C_L < 1$ be Lipschitz constant. Then, the K -hop message passing of CARIBOU satisfies:

$$\left(\frac{\alpha \Delta^2}{2 \sigma^2} \min \left\{ K, \frac{1 - C_L^K}{1 + C_L^K} \frac{1 + C_L}{1 - C_L} \right\} + \frac{\log(1/\delta)}{\alpha - 1}, \delta \right) \text{-DP}.$$

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Experiments

Accuracy: Edge-DP over the Cora dataset.

ϵ	1	2	4	8	16	32
CARIBOU	85%	87%	87%	88%	89%	89%
GAP [1]	77%	78%	77%	79%	82%	83%
DPDGC [2]	76%	78%	75%	76%	78%	80%
PertGraph [3]	60%	60%	63%	76%	85%	85%

[1] S. Sajadmanesh, A. S. Shamsabadi, A. Bellet, and D. Gatica-Perez, “Gap: Differentially private graph neural networks with aggregation perturbation,” in USENIX Security 2023.

[2] E. Chien, W.-N. Chen, C. Pan, P. Li, A. Ozgur, and O. Milenkovic, “Differentially private decoupled graph convolutions for multigranular topology protection,” in Advances in Neural Information Processing Systems, vol. 36, 2023.

[3] A. Kolluri, T. Baluta, B. Hooi, and P. Saxena, “Lpgnet: Link private graph networks for node classification,” in ACM SIGSAC Conference on Computer and Communications Security, CCS, 2022, pp. 1813–1827.

Experiments

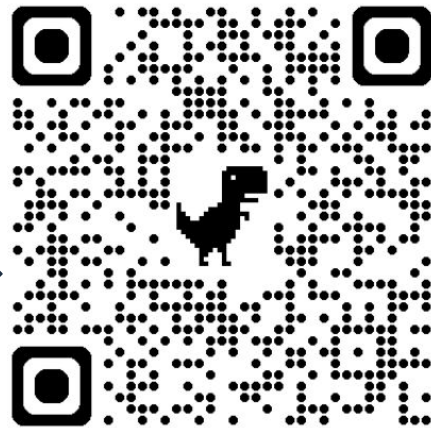
Accuracy: Node-DP over the Cora dataset.

ϵ	1	2	4	8	16	32
CARIBOU	81%	83%	86%	87%	88%	88%
GAP	34%	32%	32%	44%	56%	64%
DPDGC	34%	34%	33%	32%	28%	30%
PertGraph	19%	20%	22%	26%	28%	30%

Conclusion

- A novel privacy analysis for GNNs that leverages the contractiveness of message-passing operations to achieve convergent privacy costs.
- The design of perturbed CGL and a practical differentially private GNN framework – CARIBOU.

Full-version
Paper



Code

